

$$\begin{aligned}
& \frac{1}{6} m_c^2 c_Y^2 \overline{y}_u^{i2i3} y_u^{i1i4} + \frac{1}{16 n_c^2} \left(\frac{1}{1296} \frac{1}{m_c^2} (27 c_Y^2 \overline{y}_d^{pr} y_d^{i1r} \overline{y}_u^{i2i3} y_u^{pi4} (-1 + s_Y^2) + c_Y^2 \right. \\
& \quad (-324 s_Y^2 \overline{y}_u^{r i3} y_u^{r i4} (\overline{y}_d^{i2p} y_d^{i1p} + 3 \overline{y}_u^{i2p} y_u^{i1p}) - 81 \overline{y}_u^{pr} \overline{y}_u^{i2i3} y_u^{pi4} y_u^{i1r} (1 + c_Y^2) - \\
& \quad 27 y_u^{i1i4} (-\overline{y}_d^{i2p} y_d^{rp} \overline{y}_u^{r i3} (-1 + s_Y^2) + 3 \overline{y}_u^{pi3} \overline{y}_u^{i2r} y_u^{pr} (1 + c_Y^2) + \\
& \quad 2 \overline{y}_u^{i2i3} (5 g_1^2 + 3 g_2^2 + 48 g_3^2 - 12 s_Y^2 \overline{y}_u^{pr} y_u^{pr})) - 14 g_1^2 \overline{y}_u^{pi3} y_u^{pi4} \delta_{i1i2} + \\
& \quad 4 g_1^2 (5 s_Y^2 \overline{y}_d^{i2p} y_d^{i1p} - 13 c_Y^2 \overline{y}_u^{i2p} y_u^{i1p}) \delta_{i3i4} \Big) - \frac{1}{12} \sum c_Y g_1^2 \frac{1}{m_c^2} \overline{y}_u^{i2i3} \\
& \quad y_u^{i1i4} (c_2 c_Y c_Y - 2 s_2 c_Y s_Y) LF_{1,0}[m_d^P] + \frac{243}{4} \sum g_1^4 LF_{3,0}[m_d^P] \delta_{i1i2} \delta_{i3i4} - \\
& \quad \frac{16}{52} \sum g_1^4 LF_{4,-1}[m_d^P] \delta_{i1i2} \delta_{i3i4} + \frac{16}{1215} \sum g_1^4 LF_{5,-2}[m_d^P] \delta_{i1i2} \delta_{i3i4} - \\
& \quad \frac{1}{2} s_Y c_Y \frac{1}{m_c^4} \overline{y}_d^{pr} y_d^{pr} \overline{y}_u^{i2i3} y_u^{i1i4} (s_2 c_Y + s_Y c_Y) LF_{1,0}[m_d^r] - \\
& \quad \frac{1}{12} \sum c_Y g_1^2 \frac{1}{m_c^4} \overline{y}_u^{i2i3} y_u^{i1i4} (c_2 c_Y c_Y - 2 s_2 c_Y s_Y) LF_{1,0}[m_e^P] + \\
& \quad \frac{4}{81} \sum g_1^4 LF_{3,0}[m_e^P] \delta_{i1i2} \delta_{i3i4} - \frac{5}{54} \sum g_1^4 LF_{4,-1}[m_e^P] \delta_{i1i2} \delta_{i3i4} + \\
& \quad \frac{16}{405} \sum g_1^4 LF_{5,-2}[m_e^P] \delta_{i1i2} \delta_{i3i4} - \\
& \quad \frac{1}{6} s_Y c_Y \frac{1}{m_c^4} \overline{y}_e^{pr} y_e^{pr} \overline{y}_u^{i2i3} y_u^{i1i4} (s_2 c_Y + s_Y c_Y) LF_{1,0}[m_e^r] + \\
& \quad \frac{1}{12} c_Y \frac{1}{m_c^4} \overline{y}_u^{i2i3} y_u^{i1i4} (-2 s_Y \overline{y}_e^{pr} y_e^{pr} (s_2 c_Y + s_Y c_Y) + \sum g_1^2 (c_2 c_Y c_Y - 2 s_2 c_Y s_Y)) LF_{1,0}[m_l^P] + \\
& \quad \frac{1}{81} \sum g_1^4 LF_{3,0}[m_l^P] \delta_{i1i2} \delta_{i3i4} - \frac{5}{108} \sum g_1^4 LF_{4,-1}[m_l^P] \delta_{i1i2} \delta_{i3i4} + \\
& \quad \frac{16}{405} \sum g_1^4 LF_{5,-2}[m_l^P] \delta_{i1i2} \delta_{i3i4} - \frac{1}{12} c_Y \frac{1}{m_c^4} \overline{y}_u^{i2i3} y_u^{i1i4} \\
& \quad (6 s_Y \overline{y}_d^{pr} y_d^{pr} (s_2 c_Y + s_Y c_Y) + 6 \overline{y}_u^{pr} y_u^{pr} (c_Y^3 - s_2 c_Y s_Y) + \sum g_1^2 (c_2 c_Y c_Y - 2 s_2 c_Y s_Y)) \\
& \quad LF_{1,0}[m_q^P] + \frac{243}{4} \sum g_1^4 LF_{3,0}[m_q^P] \delta_{i1i2} \delta_{i3i4} - \\
& \quad \frac{5}{324} \sum g_1^4 LF_{4,-1}[m_q^P] \delta_{i1i2} \delta_{i3i4} + \frac{16}{1215} \sum g_1^4 LF_{5,-2}[m_q^P] \delta_{i1i2} \delta_{i3i4} + \\
& \quad \frac{1}{6} \sum c_Y g_1^2 \frac{1}{m_c^2} \overline{y}_u^{i2i3} y_u^{i1i4} (c_2 c_Y c_Y - 2 s_2 c_Y s_Y) LF_{1,0}[m_q^P] + \\
& \quad \frac{16}{243} \sum g_1^4 LF_{3,0}[m_q^P] \delta_{i1i2} \delta_{i3i4} - \frac{10}{81} \sum g_1^4 LF_{4,-1}[m_q^P] \delta_{i1i2} \delta_{i3i4} + \\
& \quad \frac{64}{1215} \sum g_1^4 LF_{5,-2}[m_q^P] \delta_{i1i2} \delta_{i3i4} - \frac{1}{2} c_Y \frac{1}{m_c^4} \overline{y}_u^{pr} \overline{y}_u^{i2i3} y_u^{pr} y_u^{i1i4} (c_Y^3 - s_2 c_Y s_Y) LF_{1,0}[m_q^r] + \\
& \quad \frac{1}{24} c_Y \frac{1}{m_c^4} \overline{y}_u^{i2i3} y_u^{i1i4} (c_Y (g_1^2 (1 - 3 c_2 c_Y^2) - 3 g_2^2 (-1 + c_2 c_Y^2)) + 3 s_4 c_Y s_Y (g_1^2 + g_2^2)) \\
& \quad LF_{1,0}[m_q] + \frac{1}{24} \frac{1}{m_c^2} c_Y^2 (3 \overline{y}_u^{i2i3} y_u^{pi4} (-s_Y^2 \overline{y}_d^{pr} y_d^{i1r} + c_Y^2 \overline{y}_u^{pr} y_u^{i1r}) - \\
& \quad y_u^{i1i4} (2 \overline{y}_u^{i2i3} (g_1^2 + 3 g_2^2) + 3 s_Y^2 \overline{y}_d^{i2p} y_d^{rp} \overline{y}_u^{r i3} - 3 c_Y^2 \overline{y}_u^{pi3} \overline{y}_u^{i2r} y_u^{pr})) LF_{1,1}[m_q] + \\
& \quad \frac{1}{216} (18 s_Y^2 c_Y^2 \overline{y}_d^{pr} y_d^{i1r} \overline{y}_u^{i2i3} y_u^{pi4} + 2 s_Y^2 \overline{y}_d^{i2p} (9 c_Y^2 \overline{y}_u^{r i3} (-6 y_d^{i1p} y_u^{r i4} + y_d^{rp} y_u^{i1i4}) + \\
& \quad 8 g_1^2 y_d^{i1p} \delta_{i3i4}) + c_Y^2 (9 \overline{y}_u^{i2i3} y_u^{i1i4} (g_1^2 + 3 g_2^2) - \\
& \quad 4 g_1^2 \overline{y}_u^{pi3} y_u^{pi4} \delta_{i1i2} - 4 \overline{y}_u^{i2p} y_u^{i1p} (27 s_Y^2 \overline{y}_u^{r i3} y_u^{r i4} + 8 g_1^2 \delta_{i3i4}))) \\
& \quad LF_{1,2}[m_q] + \frac{1}{36} (s_Y^2 \overline{y}_d^{i2p} y_d^{i1p} (9 c_Y^2 \overline{y}_u^{r i3} y_u^{r i4} - 2 g_1^2 \delta_{i3i4}) - \\
& \quad c_Y^2 (g_1^2 \overline{y}_u^{pi3} y_u^{pi4} \delta_{i1i2} + \overline{y}_u^{i2p} y_u^{i1p} (9 c_Y^2 \overline{y}_u^{r i3} y_u^{r i4} - 2 g_1^2 \delta_{i3i4}))) LF_{2,1}[m_q] + \\
& \quad \frac{1}{324} g_1^2 (9 c_Y^2 \overline{y}_u^{pi3} y_u^{pi4} \delta_{i1i2} + 2 (9 s_Y^2 \overline{y}_d^{i2p} y_d^{i1p} - 9 c_Y^2 \overline{y}_u^{i2p} y_u^{i1p} + 4 g_1^2 \delta_{i1i2}) \delta_{i3i4}) \\
& \quad LF_{3,0}[m_q] - \frac{5}{108} g_1^4 LF_{4,-1}[m_q] \delta_{i1i2} \delta_{i3i4} + \frac{405}{865} g_1^4 LF_{5,-2}[m_q] \delta_{i1i2} \delta_{i3i4} + \\
& \quad \frac{2}{81} g_1^4 LF_{3,0}[\tilde{\mu}] \delta_{i1i2} \delta_{i3i4} + \frac{1}{27} g_1^4 LF_{4,-1}[\tilde{\mu}] \delta_{i1i2} \delta_{i3i4} - \\
& \quad \frac{16}{405} g_1^4 LF_{5,-2}[\tilde{\mu}] \delta_{i1i2} \delta_{i3i4} + \frac{1}{216} g_1^2 \frac{1}{m_c^2} c_Y^2 \overline{y}_u^{i2i3} y_u^{i1i4} LF_{1,1,0}[m_1, m_q^{i1}] - \\
& \quad \frac{1}{432} g_1^2 \frac{1}{m_c^2} c_Y^2 \overline{y}_u^{i2i3} y_u^{i1i4} LF_{2,1,-1}[m_1, m_q^{i1}] + \frac{1}{1944} g_1^4 LF_{2,1,0}[m_1, m_q^{i1}] \delta_{i1i2} \delta_{i3i4} + \\
& \quad \frac{1}{1944} g_1^4 LF_{2,2,-1}[m_1, m_q^{i1}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{972} g_1^4 LF_{3,1,-1}[m_1, m_q^{i1}] \delta_{i1i2} \delta_{i3i4} + \\
& \quad \frac{1}{1944} g_1^4 LF_{4,1,-2}[m_1, m_q^{i1}] \delta_{i1i2} \delta_{i3i4} + \frac{1}{216} g_1^2 \frac{1}{m_c^2} c_Y^2 \overline{y}_u^{i2i3} y_u^{i1i4} LF_{1,1,0}[m_1, m_q^{i2}] - \\
& \quad \frac{1}{432} g_1^2 \frac{1}{m_c^2} c_Y^2 \overline{y}_u^{i2i3} y_u^{i1i4} LF_{2,1,-1}[m_1, m_q^{i2}] + \frac{1}{1944} g_1^4 LF_{2,1,0}[m_1, m_q^{i2}] \delta_{i1i2} \delta_{i3i4} + \\
& \quad \frac{1}{1944} g_1^4 LF_{2,2,-1}[m_1, m_q^{i2}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{972} g_1^4 LF_{3,1,-1}[m_1, m_q^{i2}] \delta_{i1i2} \delta_{i3i4} + \\
& \quad \frac{1}{1944} g_1^4 LF_{4,1,-2}[m_1, m_q^{i2}] \delta_{i1i2} \delta_{i3i4} + \frac{1}{27} g_1^2 \frac{1}{m_c^2} c_Y^2 \overline{y}_u^{i2i3} y_u^{i1i4} LF_{1,1,0}[m_1, m_q^{i3}] - \\
& \quad \frac{1}{27} g_1^2 \frac{1}{m_c^2} c_Y^2 \overline{y}_u^{i2i3} y_u^{i1i4} LF_{2,1,-1}[m_1, m_q^{$$